TITLE PAGE

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- **Problem Statement ID –** 26264
- **Problem statement title -** Integrated Dashboard for Monitoring and Tracking Institution-Related Complaint Cases Across AICTE Bureaus/Cells
- **Theme-** Smart automation
- **PS Category-** Software
- **Team ID-**
- **Team Name-** Nivaran-X

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AICTE UGMS-Unified Grievance Monitoring System

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AICTE OneView | Institution-centric monitoring with intelligent cross-Bureau coordination

Problem:-

Complaints handled by different AICTE bureaus may be difficult to track together, making related cases and overlapping actions harder to identify.

Proposed solution:-

A single-window, role-based dashboard consolidating institution-linked complaints across RB, PGRC and other AICTE Cells.

UGMS WORKFLOW

1 **Complaint Registration**
• capture details (id, institutions, category)

2 **Central Repository**
•store all complaints
•Standardize and organize data.

3 **Institution matching**
•find potential related records.
•filter by institution

4 **Complaint comparison**
•Compare details using AI/NLP.
•calculate similarity score

5 **Evidence based Alert**
•Flag potential duplicate
•show matching evidence

6 **Human Review**
•Verify suggestion
•Confirm /reject /defer.

Decision on Relationship

7 **Audit trail &Update**
•log action & decision
•Update status & coordinate bureaus

Innovation and key features

- Unified Dashboard: Monitor complaints across different AICTE bureaus/cells in one place.
- Smart Complaint Matching: Identify potentially duplicate or related complaints.
- Intelligent Case Classification: Distinguish related cases from different issues reported by the same institution.
- Explainable Alerts: Show why complaints may be connected to help officials review them.
- Cross-Bureau Coordination: Flag possible overlapping actions and improve coordination between bureaus.
- Human Verification: Allow officials to review suggested complaint relationships before making decisions.
- Audit Trail: Maintain records of case updates, reviews, and decisions.

TECHNICAL APPROACH

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Users

Complainants

AICTE Officials (Bureaus /cells)

Admin (System Admin)

Presentation Layer (Frontend)

- Web Portal
- Mobile Responsive
- User Dashboard
- Secure Login

Application Layer (Backend)

- API Gateway (Handles Request)
- Business Logic (Service Layer)
- Authentication & Authorization
- Notification Service
- Logging & Monitoring

AI/ML Layer (Decision Support)

- Similarly Matching (Find Related/Duplicate Complaints)
- Case Classification (Duplicate/Related/ Different Issues)
- Explainable AI (Reasons For Suggessions)

Data Layer Database & Integration

- Central Database (Complaints,Logo)
- AICTE Systems (Bureaus/cells)
- Institutions Data (Reference Information)

Output Layer (Result & Actions)

- Unified Dashboard (Real Time Tracking)
- Alert & Suggestions (Potential Connections)
- Report & Analytics Insights & Trends

TECH STACK

Frontend	React.Js, TailWind.css
Backend	Python, FastAPI
AI / ML	Skicit.Learn
Database	SQlite, PostgreSQL
Deployment	Docker, Cloud

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FEASIBILITY AND VIABILITY

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Feasibility

Can we built it.?

Accessible technologies
Uses beginner-friendly web development and text-matching tools.

Modular development
Build the dashboard, complaint matching, alerts, and review system step by step.

Explainable matching
Use institution details, keywords, and text similarity to suggest related cases.

Prototype Demonstration
Test the system using synthetic complaint data without requiring internal AICTE access.

Viability

Is it practicle and useful?

Long Term Potential
Begin with open source tool & minimal infrastructure

Scalable Design
Start with small prototype and plan for larger dataset

Operational Value
Help officials identify potentials related complaints across bureuas

Long Term Potential
Support better co-ordination,traceability and complaint monitoring

Challenges & Risks	Possible Solutions
Different Institution Names — The same institution may appear under different names or abbreviations.	Use institution IDs, name normalization, and alias matching.
Incorrect Case Matching — Unrelated complaints may appear similar.	Display matching evidence and require human verification before linking cases.
Data Privacy & Security — Complaint records may contain sensitive information.	Use synthetic data for the prototype and implement role-based access controls.
Large Complaint Volumes — Comparing many complaints may increase processing time.	Filter by institution, category, and date before comparing relevant cases.

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IMPACT & BENEFITS

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Economic Impact

- **Reduced Administrative Effort:** Minimizes repetitive manual complaint checking.
- **Cost-Effective Operations:** Uses open-source technologies for the initial prototype.
- **Better Resource Utilization:** Helps officials prioritize cases requiring coordination.
- **Less Duplicate Work:** Helps identify potentially duplicate reviews across bureaus.

Social Impact

- **Improved Grievance Monitoring:** Makes institution-related complaints easier to track.
- **Better Coordination:** Helps bureaus identify cases that may require joint attention.
- **Potentially Better Service:** Supports more organized complaint handling for students and institutions
- **Greater Transparency:** Maintains traceable records of case-linking decisions.

Environmental Impact

- **Paper-Efficient Administration:** Supports a more paper-efficient complaint-monitoring workflow.
- **Reduced Paperwork:** Encourage digital complaint record instead of paper-based tracking.
- **Less Printing:** Enables digital review and reporting.

Technical Impact

- **Unified Dashboard:** Brings complaint information from different bureaus into one view.
- **Intelligent Case Matching:** Uses institution details and text similarity to identify potential relationships.
- **Explainable Alerts:** Shows evidence behind suggested case links.
- **Auditable Decisions:** Records review outcomes and case updates.

Centralized Complaint Monitoring

- •One dashboard for complaint monitoring.
- •Easier access to related case information.
- •Centralized view of complaint status and updates.

Faster Case Identification.

- •Intelligent complaint matching.
- •Explainable alerts for possible case relationships.
- •Highlights potentially duplicate or overlapping cases.

Improved Inter-Bureau Coordination

- •Flags cases requiring joint attention.
- •Helps officials review related complaints together.
- •Supports coordinated follow-up and action tracking.

Transparency and Accountability.

- •Human verification before linking cases.
- •Audit trail of reviews and decisions.
- •Recorded complaint updates and review outcomes.

RESEARCH AND REFERENCES

Feature	Existing AICTE System	Proposed UGMS
Grievance Submission	✓ Centralized Support System	Uses authorized complaint data
Bureau/Cell Routing	✓ Routes queries to concerned officials	Complements existing routing
Complaint Processing	✓ Existing grievance mechanisms	Supports related-case monitoring
Unified Cross-Bureau View	Not verified	✓ Unified monitoring dashboard
Related-Case Identification	Not verified	✓ Institution and text-based matching
Potential Duplicate Alert	Not verified	✓ Flags potentially related complaints
Coordination Support	Existing bureau/cell workflows	✓ Highlights cases needing joint attentio
Case-Link Verification	Existing official review procedures	✓ Human verification before linking
Review Audit Trail	Not verified	✓ Records case-link decisions and update
Analytics & Reporting	Existing AICTE dashboards are available	✓ Proposed related-case insight and report

REFERENCE	YEAR	LINK
AICTE Approval Process Handbook-Complaint Cases	2024	Link
AICTE(Establishment of Mechanism For Grievance)Guidelines	2012	Link
AICTE(Redressal Of Grievance Of Student)Regulations	2019	Link
AICTE Grant of Approval for Technical Institutes	2020	Link
AICTE Centralized Support System-Grievance And Query Submission	Official Portal	Link